19 November 1969
Superseding
MIL-A-55674(EL)
27 August 1969

MILITARY SPECIFICATION

ANTENNA GROUP AN/TRA-37( )

1. SCOPE

1.1 This specification covers the characteristics and requirements for the fabrication and testing of Antenna Group AN/TRA-37( ) with it's associated ancillary items. The antenna group is primarily used in tactical applications since it is easily transportable. Antenna Group AN/TRA-37( ) is for use with radio equipment employing tropospheric scatter mode of transmission up to 2 kw RF power in the frequency range of 4.4 to 5.0 GHz. (See 6.1 and 6.6).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GGG-H-86	Hammers, Hand and Maul
CCC-C-419	Cloth, Duck, Cotton, Un-Bleached, Plied-yarns, Army and Numbered
L-P-523	Plastic Sheet and Film, Fep-Fluorocarbon, Extruded
PPP-B-621	Box, Wood, Nailed and Locked Corner
PPP-B-636	Box, Fiberboard
ZZ-R-765	Rubber, Silicone, Low and High Temperature and Tear Resistant
QQ-S-781	Steel Strapping, Flat
PPP-C-843	Cushioning, Material Cellulosic.
PPP-C-850	Cushioning Material, Polystyrene Expanded, Resilient for Packaging Uses

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T-T-871	Twine, Cotton, Wrapping
CCC-D-950	Dyeing and after treating Process for Cotton Cloths

MILITARY

MIL-P-116	Preservation, Methods of
MIL-R-3065	Rubber, Fabricated Parts
MIL-C-3133	Cellular Elastomeric Materials, Fabricated Parts
MIL-S-6855	Synthetic Rubber Sheets, Strips, Molded or Extruded Shapes
MIL-R-7362	Rubber, Synthetic, Solid, Sheet and Fabricated Parts, Synthetic Oil Resistant
MIL-P-10971	Pins, Spring, Tubular, Coiled and Slotted
MIL-P-11268	Parts, Materials and Processes used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-C-13924	Coating Oxide, Black, For Ferrous Materials
MIL-F-14072	Finish for Ground Signal Equipment.
MIL-P-20700	Pins, Grooved, Headless, Longitudinal Groove
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging
MIL-L-46010	Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting

STANDARDS

FEDERAL

FED-STD-595	Color, Requirements for Individual Color Chips

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

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MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-417	Rubber Compositions, Vulcanized General Prupose, Solid
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing with appropriate Test Methods
MS-51926-3	Clip End, Strap Type
MS-51929-2	Buckle, Spring Action

MILITARY HANDBOOK

MIL-HDBK-132	Military Handbook Protective Finishes

DRAWINGS

ELECTRONICS COMMAND

SC-B-61956	Stake, GP-2
DL-SM-B-655450	Antenna Group AN/TRA-37
SC-A-33076	Metals Guide
SC-D-459376	Anchor Assembly "W" Type
SC-A-46420	Fasteners
SC-A-47792	Loop, Footman
SC-B-539505	Hammer, Sledge

(Copies of specifications, standards, handbooks, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, 25, D.C.)

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3. REQUIREMENTS

3.1 Description. - Antenna Group AN/TRA-37( ) consists of two each solid thin-skinned, 10 foot diameter parabolic antenna assemblies installed on 15 foot masts with ancillary equipment and hardware. The group shall provide for tropospheric scatter transmission of 2 KW RF power and space diversity reception in the frequency range of 4.4 to 5.0 GH_Z.

3.1.1 Components. - Antenna Group AN/TRA-37( ) (SM-F-655451) consists of the following major components:

Two (2) each Antenna, AS-1804( )/TRA-37 (DL-SM-B-655458)

Two (2) each Mast, AB-1083( )/TRA-37 (DL-SM-B-655456)

Installation Tool Set (PL-SM-B-655483)

Erection Equipment Set (PL-SM-B-655482)

Assembly, Flexible Waveguide (REC. A) (SM-C-655507)

Assembly, Flexible Waveguide (REC. B) (SM-C-655571)

Assembly, Flexible Waveguide (XMIT) (SM-C-655572)

Two (2) each Feedhorn, Space Diversity (SM-D-655492)

Reel, Storage, Flexible Waveguide (SM-D-655486)

3.2 First article samples (preproduction samples). - The contractor shall furnish two (2) each Antenna Group AN/TRA-37( ) first article samples for approval, unless otherwise noted in the invitation for bids and contract.

3.2.1 Fabrication and assembly. - First article samples shall be fabricated by the contractor and shall meet all requirements specified for the equipment on contract. The first article samples shall be assembled using parts, materials, and processes that will be employed in production, and shall be fabricated and assembled in a manner similar to that to be used in production.

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3.2.2 Approval and use. - Approval of first article samples shall not be construed as a waiver of any specified requirement. After being releases to the contractor (See 3.2.3), first article samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.2.3 Reference standards. - After first article samples have been approved, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.3 Construction. - The equipment shall be constructed in accordance with the requirements of this specification, and of Drawing List DL-SM-B-655450, (Including the requirements for parts, materials, and processes thereon). In case of conflict between this specification and the drawings, the requirements of this specification shall govern.

3.4 Parts, Materials and Processes; General. - In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.5 Fibrous material, organic. - Organic fibrous material shall conform to requirement 44 of MIL-STD-454, except that, when duck is not required to be fire resistant,it shall conform to CCC-C-419 and shall be given class B after treatment, using inhibitors "a", "c", "f", or "g" as specified in CCC-D-950. Inhibitor "a" shall be used when the duck will come in contact with natural or synthetic rubber or prolonged contact with skin can occur. The color of the fabric shall be olive-drab No. 7. The color of thread shall match shade S-1 of the United States Army Standard Color Card for Official Standardization Shades of Sewing Threads. (See 6.7)

3.6 Lubricants. - Dry lubricant (baked on all bearing surfaces) shall be compound RIA-9A in accordance with MIL-L-46010 as made by Sandstrom Products Co., Port Byron, Illinois, or equal. Note: Graphite, molydenum disulfide, or sulfonates containing chloride shall not be used in lubricant or anti-sieze compounds.

3.7 Metals. -

3.7.1 General. - Drawing SC-A-33076 shall be consulted as a guide in the selection of metal specifications for particular applications; however, first consideration shall be given to other specified requirements, including requirements on overall equipment performance and service conditions. Where dissimilar metals are in contact, they shall be protected as required by MIL-F-14072.

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3.7.2 Castings. - Castings shall conform to requirement 21 of MIL-STD-454. Zinc-alloy, aluminum-alloy, and magnesium-alloy die castings shall be used only when approved by the contracting officer for the particular application.

3.8 Thermoplastic. - Thermoplastic material and parts shall conform to L-P-523. These materials shall not be treated and shall be masked, if necessary, during any treatment of the equipment shich might result in degradation of the electrical and physical characteristics of the material. Only fungistatic compositions in accordance with requirement 4 of MIL-STD-454 shall be used or compounds modified with fungicides as an ingredient and capable of passing the fungus test, Method 508, of MIL-STD-810.

3.9 Rubber. - Only fungus-resistant grades in accordance with requirement 4 of MIL-STD-454 shall be used. Rubber (natural or synthetic) shall conform to MIL-C-3133 for cellular type, MIL-R-7362 for rubber resistant to diester synthetic oils, ZZ-R-765 for silicone rubber, and MIL-R-3065 and MIL-STD-417 for other types.

3.9.1 Rubber Reflector Bumpers. - The rubber reflector bumpers shall meet the requirements of MIL-S-6855, Class 2.

3.10 Cleaning. -

3.10.1 Parts. - After fabrication, parts shall be cleaned in accordance with requirement 9 of MIL-STD-454, or as specified in an applicable document.

3.10.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, such cleaning shall also be performed before final assembly of the equipment.

3.11 Finish, protective. - Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware type items, such as handles, hinges, screws, etc, and necessary touch up after mounting. Unless otherwise specified, the final paint film on type I surfaces (See MIL-F-14072), exposed to view during operation, shall be a smooth, semigloss (Olive drab) green, in color conforming to No. X24087 of FED-STD-595. When some other color or gloss is referred to in the equipment specification, but the details are not specified, the contractor shall apply to the contracting officer for information regarding the required finish. For interference reduction purposes, interface surfaces of structures and bonding connections between cases and mounting bases shall have a finish for electrical contact. (See 6.3)

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3.11.1 Finish, stainless steel. - All exterior stainless steel assemblies and hardware shall be black oxide coated conforming to specification MIL-C-13924.

3.11.2 Reflective Hardware. - All other bright, shiny metal hardware such as buckles, screws, bolts, chains, etc., shall be given a suitable non-reflective finish in accordance with MIL-F-14072 or MIL-HDBK-132.

3.12 Marking of equipment. - Marking shall conform to MIL-M-13231. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.12.1 Serial numbers. - The following items shall have serial numbers: All nomenclatured units.

3.12.2 Identification Marking. - Equipment identification marking shall be made in accordance with MIL-STD-130.

3.13 Riveting. -(See MIL-P-11268)

3.14 Securing of parts. - (See MIL-P-11268)

3.15 Soldering and brazing. - (See MIL-P-11268)

3.16 Tropicalization. -

3.16.1 Treatment of materials. - Fungus-inert materials shall be used where practicable. These materials shall be in accordance with Requirement 4 of MIL-STD-454. When materials are required to be treated, they shall be cleaned in accordance with Requirement 9 of MIL-STD-454 and then shall be treated in accordance with 3.16.2 through 3.16.5 and as follows:

Fibrous material, organic (See 3.5)

3.16.2 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.16.3 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the items in which the treated material is used, or during use of the finished items when used for the purpose intended.

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3.16.4 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.16.5 Statement of treatment. - The statement of treatment shall indicate whether or not nutrient materials have been used. If such materials have been used, it shall also describe in detail the materials to be treated and the treating materials and processes that the contractor proposes to use. (See 6.2)

3.17 Nonstandard parts. - When nonstandard parts must be used, such parts shall be capable of passing the moisture resistance test, Method 106 of MIL-STD-202; the fungus test, Method 508 of MIL-STD-810, 28 days minimum, without evidence of fungus growth on any surfaces; and 48 hours of salt-fog test, Method 509 of MIL-STD-810 without evidence of corrosion.

3.18 Structural welding. - (See MIL-P-11268)

3.18.1 Cleaning, prior to welding. - (See MIL-P-11268)

3.18.2 Process. - (See MIL-P-11268)

3.18.3 Cleaning after welding. - (See MIL-P-11268)

3.19 Bearings. - (See MIL-P-11268)

3.20 Controls. - Controls shall conform to Requirement 28 of MIL-STD-454.

3.21 Gear assemblies, gears and cams. - (See MIL-P-11268)

3.22 Hardware, miscellaneous. - Hardware for canvas items shall conform to MS-51926-3, MS-51929-2 and SC-A-47792.

3.23 Level vials. - (See MIL-P-11268)

3.24 Lubrication fittings. - (See MIL-P-11268)

3.25 Screws, other threaded devices, and related parts. - Screws, other threaded devices, and related parts shall conform to Requirement 12 of MIL-STD-454, except as otherwise specified in MIL-P-11268.

3.26 Spring and grooved pins. - Spring pins conforming to MIL-P-10971 or grooved pins conforming to MIL-P-20700 shall be used where taper pins would normally be employed.

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3.27 Waveguides, waveguide assemblies, and related items. - (See MIL-P-11268).

3.28 Overall characteristics. - Antenna Group AN/TRA-37( ) shall have the following features and characteristics:

3.28.1 Antenna Gain. - The gain of the antenna relative to an isotropic radiator shall be a minimum of 40 db, -20 log 5000/f MHz for the frequency band of 4.4 to 5.0 GHz (f is the frequency in MHz). (See 4.15)

3.28.2 Beamwidth. - The 3db beamwidth shall be 1.5^{o} (plus or minus 0.75^{o} under all operating conditions). (See 4.15).

3.28.3 Minor Lobes. - The ratio of major to any minor lobe shall be 19 db minimum at any frequency in the 4.4 to 5.0 GHz band for any polarization. (See 4.15)

3.28.4 Overall Antenna VSWR. - The VSWR shall be no greater than 1.58 over the frequency range of 4.4 to 5.0 GHz when measured at the input of the 25 foot flexible waveguide section. (See 4.15).

3.28.5 Overall Antenna Isolation. - At any frequency in 4.4 to 5.0 GHz range and for all antenna adjustments, there shall be a minimum of 40 db coupling loss between the transmitter and receiver sections. The coupling loss shall be measured at the receiver termination of 25 feet of transmission line with a signal applied at the transmitter termination of 25 feet of transmission line. (See 4.15)

3.28.6 Transmission Line Attenuation. - The transmission line attenuation shall not exceed 1 db/25 feet over the frequency range of 4.4 to 5.0 GHz. (See 4.15.1)

3.28.7 Antenna Feedhorn. - The antenna feedhorn shall be so constructed that external accumulation of dust, sand, salt deposits, moisture and ice will cause no degradation in performance. Pressurization of the feedhorn during operation shall not be necessary, however, it shall meet the Watertightness requirements of 3.28.7.3 herein. It shall be supported by the reflector and spar assembly independent of the mast. The feedhorn shall have the following characteristics:

3.28.7.1 Feedhorn VSWR. - The VSWR as measured at the input of the transmit port and the receive port shall not exceed 1.35:1, at 4.4, 4.7 and 5.0 GHz. (See 4.11)

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3.28.7.2 Feedhorn Coupling Loss (Isolation). - The coupling loss as measured at the receive port with a signal applied at the transmit port shall not be less than 40 db, at 4.4, 4.7 and 5.0 GHz (See 4.12).

3.28.7.3 Feedhorn Watertightness. - The feedhorn shall be watertight and the leakage, if any, shall be less than 0.25 cubic feet per hour to insure watertight construction. (See 4.13).

3.28.7.4 Feedhorn High Power Capability. - The feedhorn assembly shall exhibit no evidence of corona or arc breakdown under power of 2 KW (± 400 W) in the frequency range 4.4 to 5.0 GHz. (See 4.14)

3.28.7.5 Feedhorn Polarization. - The feedhorn shall be mounted such that the plane or polarization of the transmitted signal shall be normal to the plane of polarization of the received signal, and each shall be at angle of 45 degrees with respect to horizontal. In addition, provisions shall be made to provide the capability of operation in a conventional 0-90° polarization relative to the ground for non-standard usage.

3.28.8 Azimuth adjustment. - The antenna shall be adjustable in azimuth through a range of plus or minus 30 degrees continuous. Control shall be provided to permit vernier movement of the antenna. The azimuth scale or readout device shall indicate bearings in the range of 0 to 360 degrees and shall have the capability of being zeroed to an established reference. Markings are to be in at least 0.5 degree increments. All adjustments must be made from ground level. The azimuth control lever torque under assembled conditions shall be a maximum of 75 inch pounds. (See 4.16)

3.28.9 Elevation Adjustment. - The antenna shall be adjustable in elevation from minus 5 degrees to plus 15 degrees off horizontal, referenced to true vertical. Scale markings shall be in increments of 0.2 degrees referenced to horizontal.

3.28.10 Sighting mechanism. A sighting mechanism shall be provided with each antenna so that the antenna may be oriented in the desired direction.

3.28.11 Gin pole. - The gin pole shall provide the capability for erecting the antenna and rotating the antenna after erection.

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3.28.12 Mast base. - The antenna mast base shall contain a ball joint to permit alignment on terrain having a slope up to 15 degrees. The baseplate assembly shall be designed to permit a mast depression angle of 90 degrees from zenith to allow the mast to rest on the ground during the assembly procedure.

3.28.13 Antenna Reflector. - The reflector shall consist of four (4) interchangeable quadrants, with captive hardware to secure radial joints, peripheral connections, reflector to hub attachment, and spar connections. The hardware required for the attachment of the spar assembly shall be contained in each reflector quadrant. The reflector shall contain attachment provisions to accept the reflector yoke assembly.

3.28.13.1 Reflector shape, size and weight. - The antenna reflector shall be parabolic in shape, ten (10) feet (+ 1/2 inch) in diameter with a .40 f/D Ratio. The weight shall not exceed 150 pounds.

3.28.13.2 Reflector Contour. - The reflector surface shall not deviate from the required theoretical contour by more than 0.048 inch RMS. The reflector must withstand a windload of 50 mph with 1/2 inch of radial ice (Front and Rear) with a total deformation (Including the initial contour deviation) not to exceed 0.080 RMS, after allowing for a maximum angular correction of 10 minutes of arc for contour rotation.

3.28.13.3 Interface Accuracy. - Angular misalignment between the Feedhorn Assembly mounting surface on the reflector yoke assembly and the plane of the front edge of the reflector shall not exceed 0.15° degrees.

3.28.14 Transit Assembly. - A pocket transit is supplied with each AN/TRA-37 as an aid in sighting during erection and installation of the equipment. The transit contains as a minimum, a graduated azimuth scale, a round level, an elevation vernier, a long level for use in sighting elevation angles, and a compass needle.

3.28.15 Sledge Hammer, p/o Installation Tool set. - One (1) each Sledge Hammer shall be provided as part of the Installation Tool Set supplied with each Antenna Group AN/TRA-37( ). The Hammer shall be Style A, Type X, Class 1, 8 pound, per Specification GGG-H-86.

3.29 Service Conditions.- Each configuration shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in any combination to be encountered. Each unit shall meet handling and servicing conditions without damage during installation and maintenance.

3.29.1 Temperature tests. -

3.29.1.1 High and Low. - During testing as specified in 4.6.1 and 4.6.2, the equipment shall meet the following requirements: All rotatable

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and otherwise adjustable devices, specifically, the azimuth and elevation adjustment controls, the winch and crank, and the tensioning nuts on the guy cable take-up devices shall be found to be mechanically operational (free and smooth) in accordance with this specification. In addition, both Feedhorn VSWR in accordance with 3.28.7.1 and Coupling Loss in accordance with 3.28.7.2 herein shall be checked at 4.4, 4.7 and 5.0 GHz.

3.29.2 Humidity (Moisture-Resistance) Test. - The equipment shall be capable of successful operation at high relative humidity up to 100% at all temperatures from -65°F to + 85°F and with condensation from +33°F, and relative humidities equivalent to 13 grains of moisture per cubic foot of air at all temperatures between 95°F and 125°F and low relative humidity of 5% at a temperature of + 125°F. During testing as specified in 4.7, all rotatable and otherwise adjustable devices shall be found to be mechanically operational in accordance with this specification. In addition, both the Feedhorn VSWR in accordance with 3.28.7.1 and Coupling Loss in accordance with 3.28.7.2 herein, shall be checked at 4.4, 4.7 and 5.0 GHz.

3.29.3 Wind and Ice Load. -

3.29.3.1 Wind Load. - The Antenna Group erected shall be capable of operating in winds of 50 mph continuous and survive in winds of 100 mph (42.0 psf) peak, no ice, in any direction without permanent deformation. (See 4.8)

3.29.3.2 Ice Load, Operating. - The Antenna Group erected shall be capable of operating with 1/2 inch (4.5 lbs/sq. ft.) of radial ice in peak winds of 75 mph (23.6 psf), in any direction without permanent deformation. (See 4.8)

3.29.3.3 Ice Load, Non-Operating. - Permanent deformation shall not occur in any of the antenna group components while stored in open transport frames covered with one (1) inch of radial ice and subjected to 75 mph peak winds.

3.29.3.4 Operation. - Under all operating conditions for Wind and Ice-Load characteristics (See 3.29.3.1 and 3.29.3.2), all rotatable and otherwise adjustable devices shall be mechanically operational in accordance with this specification. The total azimuth deflection (twist) of the antenna shall not exceed plus or minus $\pm$ 0.6° degrees. In addition, the change in Antenna Gain both before, and during wind and ice loading conditions shall not be more than 3 db (+ 2db, -any). Testing shall be accomplished per 4.8 and 4.8.1.

3.29.3.5 Survival. - After subjection to the maximum frontal and rear wind and ice load conditions, the change, in Antenna Gain both before and after testing shall not vary by more than 1.0 db. All rotatable and otherwise adjustable devices shall be mechanically operational in accordance with this specification. (See 4.8.1)

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3.29.4 Vibration. - Vibration tests shall be conducted only on the Antenna Feedhorn, Upper Mast Section (AB-1084/TRA-37), and Center Mast Section (AB-1085/TRA-37). Following testing as specified in 4.9.1 and 4.9.2, all rotatable and otherwise adjustable devices shall be found to be mechanically operational in accordance with this specification. In addition, the Feedhorn VSWR shall meet the requirements of 3.28.7.1 and Watertightness requirements of 3.28.7.3 herein.

3.29.5 Altitude (Elevation). - The Feedhorn shall meet full specification performance for the following measurements both during and after testing as specified in 4.10. Feedhorn VSWR shall meet the requirements of 3.28.7.1, and the feedhorn shall exhibit no evidence of corona or arc-over (See 3.28.7.4) under RF power of 2 KW $\pm$ 400 W during operation at frequency settings of 4.4, 4.7 and 5.0 GH_Z at an altitude of 10,000 feet (20.6 in. of Hg).

3.30 Mechanical Interchangeability. - Like units, assemblies, sub-assemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand-picked for fit or performance. The equipment shall be tested and inspected in accordance with 4.17 herein.

3.31 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of paragraphs 3.5 through 3.30 of this specification. (See 4.18)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facility or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. - Inspection shall be classified as follows:

(a) First Article inspection (preproduction), (does not include preparation for delivery). (See 4.3)

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(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5).

(2) Quality conformance inspection of preparation for delivery. (See 4.19).

4.3 First article inspection. - This inspection (See 4.2) shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B and Group C. (See Tables III, IV and V respectively). The first article inspection shall be performed in the order shown in Table I. (NOTE: Each Antenna Group AN/TRA-37 consists of two (2) each antenna systems (two masts, two feedhorns, two antennas, and a dual compliment of ancillary items) and a single set of erection equipment tools. Therefore, unit numbers in Table I denote half of an Antenna Group less erection set and tools.) Table II details the order of inspection for Service Condition testing performed on the two sets of erection equipment and tools which are part of the two sample antenna groups.

TABLE I
FIRST ARTICLE INSPECTION AND
ORDER OF INSPECTION FOR
SERVICE CONDITION TESTING
(ANTENNA AND MAST EQUIPMENTS)

Inspection	Req Para	Insp Para	Number of Samples-Two			
			Unit 1	Unit 2	Unit 3	Unit 4
Temperature-High	3.29.1.1	4.6.1			1	
Temperature-Low	3.29.1.1	4.6.2			2	
Altitude (Feedhorn only)(Elevation)	3.29.5	4.10	2			1
Vibration (Feedhorn & Mast Secs. only, See Insp para)	3.29.4	4.9.1 4.9.2		1		2
Humidity	3.29.2	4.7			3	
Wind & Ice Load	3.29.3	4.8	1			

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TABLE II
FIRST ARTICLE INSPECTION AND
ORDER OF INSPECTION FOR
SERVICE CONDITION TESTING
(ERECTION AND TOOL SET EQUIPMENT)

Inspection	Req Para	Insp Para	Sample 1	Sample 2	
Temperature - High	3.29.1.1	4.6.1	1	1	
Temperature - Low	3.29.1.1	4.6.2	2	2	
Humidity	3.29.2	4.7	3	3	

4.4 Inspection Covered by Subsidiary Documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery.(See 4.2)

Item	Where Required
Parts, Materials and Processes	3.4
Finish	3.11
Marking	3.12
Workmanship	3.31

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.5 and 4.6.1 through 4.6.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government, at its discretion, may perform all or any part of the specified inspection, to verify the

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contractor's compliance with specified requirements. (See 6.8) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A Inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table III. Each lot will be subjected to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in Table III. Group A inspection shall be performed in any order which is satisfactory to the Government, except that operational (electrical) inspection, shall be performed last.

TABLE III
GROUP A
VISUAL, MECHANICAL & ELECTRICAL INSPECTION

Inspection Item	Drawings	AQL		Req Para	Insp Para
		Major	Minor		
Antenna Group AN/TRA-37 c/o	DL-SM-B-655450				
Antenna AS-1804/TRA-37 c/o	DL-SM-B-655458				
Reflector, Sectionalized Paraboloid	SM-D-655474	1%	4.0%	3.31	4.18
Yoke, Reflector	SM-D-655491	1%	4.0%	3.31	4.18
Feedhorn, Space Diversity	SM-D-655492	1%	4.0%	3.31	4.18
Assembly, Spar	SM-D-655501	1%	4.0%	3.31	4.18
Assembly, Cable-Waveguide Support	SM-D-655493	1%	4.0%	3.31	4.18
Mast AB-1083/TRA-37 c/o	DL-SM-B-655456				

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Table III, Group A (cont'd)

Inspection Item	Drawings	AQL Major	AQL Minor	Req Para	Insp Para
Mast Section, Upper AB-1084/TRA-37	DL-SM-B-655459	1%	4.0%	3.31	4.18
Mast Section, Center AB-1085/TRA-37	DL-SM-B-655460	1%	4.0%	3.31	4.18
Mast Section, Lower AB-1086/TRA-37	DL-SM-B-655461	1%	4.0%	3.31	4.18
Base, Mast AB-1087/TRA-37	SM-D-655478	1%	4.0%	3.31	4.18
Guy, Upper w/Take-up Device	SM-D-655506	1%	4.0%	3.31	4.18
Guy,Lower w/Take-up Device	SM-D-655485	1%	4.0%	3.31	4.18
Anchor Assy "W" Type	SC-D-459376	1%	4.0%	3.31	4.18
Stake, GP-2	SC-B-61956	1%	4.0%	3.31	4.18
Installation Tool Set c/o	PL-SM-B-655483				
Trenching Tool	SM-B-655489	1%	4.0%	3.31	4.18
Hammer, Sledge	SC-B-539505	1%	4.0%	3.31	4.18
Transit Assy	SM-C-655591	1%	4.0%	3.31	4.18
Erection Equipment Set c/o	PL-SM-B-655482				
Gin Pole	SM-D-655487	1%	4.0%	3.31	4.18
Cable, Erection-Gin Pole	SM-C-655494	1%	4.0%	3.31	4.18
Cable, Stabilizing Gin Pole	SM-C-655502	1%	4.0%	3.31	4.18

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Table III, Group A (cont'd)

Inspection Item	Drawings	AQL Major	AQL Minor	Req Para	Insp Para
Assembly, Winch	SM-B-655509	1%	4.0%	3.31	4.18
Support Assy, Mast (BIPOD)	SM-C-655625	1%	4.0%	3.31	4.18
Cable, Mast Support	SM-D-655495	1%	4.0%	3.31	4.18
Cable-Base Plate, Tie Back	SM-C-655503	1%	4.0%	3.31	4.18
Cable, Erection-Truck Assist	SM-C-655496	1%	4.0%	3.31	4.18
Assembly, Flexible Waveguide (Receive A)	SM-C-655507	1%	4.0%	3.31	4.18
Assembly, Flexible Waveguide (Receive B)	SM-C-655571	1%	4.0%	3.31	4.18
Assembly, Flexible Waveguide (Transmit)	SM-C-655572	1%	4.0%	3.31	4.18
Reel, Storage, Flexible Waveguide	SM-D-655486	1%	4.0%	3.31	4.18

Electrical Inspection	Req Para	Insp Para	AQL*
Feedhorn, VSWR	3.28.7.1	4.11	1.0%
Feedhorn, Isolation	3.28.7.2	4.12	1.0%

*NOTE: All Electrical Defects Are Major.

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4.5.2 Group B Inspection. - This inspection, including sampling, shall conform to Table IV and to the special procedures for small sample inspection of MIL-STD-105. The AQL shall be 6.5 percent defective and Inspection Level shall be S-4 for Normal and Tightened Inspection and Level S-3 for Reduced Inspection. Group B Inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met Group A inspection.

TABLE IV
GROUP B - Mechanical, Electrical and Environmental Inspection

Inspection	Req Para	Insp Para	AQL
Antenna Gain	3.28.1	4.15	6.5% - Level
Beamwidth	3.28.2	4.15	S-4
Minor Lobes	3.28.3	4.15	"
Overall Antenna VSWR	3.28.4	4.15	"
Overall Antenna Isolation	3.28.5	4.15	"
Feedhorn Watertightness	3.28.7.3	4.13	"
Feedhorn High Power Capability	3.28.7.4	4.14	"
Vibration - Part I, Feedhorn Only	3.29.4	4.9.1	"
Altitude (Elevation) (Feedhorn Only)	3.29.5	4.10	"
Mechanical Interchangeability	3.30	4.17 (Note 1)	"

NOTE 1: Inspection and testing may be accomplished in any order which is deemed practicable, however, mechanical interchangeability tests shall be performed on complete Antenna Groups only.

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4.5.3 Group C Inspection. - This inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and met Group A and Group B Inspections. Each sample unit used for the vibration test need meet only those environmental group C inspections not indicated for that unit in Table IV. Sample units shall be selected in accordance with paragraph 4.5.3.1.

TABLE V
GROUP C INSPECTION

Inspection	Req Para	Insp Para
Group I (1/2 Antenna Group)		
High and Low Temperature Test	3.29.1.1	4.6.1 4.6.2
Humidity (Moisture Resistance Test)	3.29.2	4.7
Vibration Test (Part 2 - Mast Sections)	3.29.4	4.9.2
Group II (1/2 Antenna Group)		
Wind and Ice Load Tests	3.29.3	4.8 4.8.1
Wind Load	3.29.3.1	4.8, 4.8.1
Ice Load, Operating	3.29.3.2	4.8, 4.8.1
Ice Load, Non-Operating	3.29.3.3	4.8, 4.8.1
Operation	3.29.3.4	4.8, 4.8.1
Survival	3.29.3.5	4.8, 4.8.1

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4.5.3.1 Sampling for inspection. - One (1) sample antenna group shall be selected at random for all inspections, as applicable in Table V every two months (60 days) and fraction thereof produced. The first sample selected shall be at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3).

4.5.3.2 Noncompliance. - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted: however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective. (See 6.5)

4.5.4 Reinspection of Conforming Group C Sample Units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Temperature Test. -

4.6.1 High temperature. - The equipment shall be tested in accordance with Method 501, Procedure I, of MIL-STD-810. The storage temperature(which includes effect of sun-load) shall be + 160°F (+6° -0)°F. The operating temperature shall be +125°F (+6° -0)°F. Prior to initiation of testing at a temperature of 77°F ± 3°F @ Rel. Hum. < 75% and during testing at a temperature of +125°F, the equipment shall meet full specification performance for the mechanical operations and electrical measurements specified in 3.29.1.1.

4.6.2 Low temperature. - The equipment shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be -80°F (+0, -6)°F. Maintain the equipment at the extreme low range for a minimum period of 24 hours or to practical thermal equilibrium, whichever occurs first. The operating temperature shall be -65°F (+0, -6)°F. Prior to initiation of testing at a temperature of 77°F ± 3°F @ Rel. Hum. < 75% and during testing at a temperature of - 65°F, the equipment shall meet full specification performance for the mechanical operations and electrical measurements specified in 3.29.1.1

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4.7 Humidity (Moisture-Resistance) Test. - The equipment shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to initiation of testing and once (1) every twenty-four (24) hours thereafter, until completion of the entire five (5) 48 hour cycles (See Figure 507-2 in MIL-STD-810) the equipment shall meet full specification performance for the mechanical operations and electrical measurements specified in 3.29.2. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

4.8 Operating, Wind and Ice Loading test. - The equipment shall be tested for the wind and ice load requirements of 3.29.3.1, 3.29.3.2 and 3.29.3.4 herein.

4.8.1 Test Method. - A simulated Ice and Wind Load condition through use of sand bags and guy tensioning devices with tensiometer may be used by the contractor for performance of this test, however, any proposed test method to be used shall be submitted to the responsible production engineering commodity division for approval prior to it's use.

4.9 Vibration. -

4.9.1 Vibration - Part 1 (Resonance Search). - Mast Section AB-1084/TRA-37, Mast Section AB-1085/TRA-37 and the Antenna Feedhorn shall be subjected separately to the vibration tests prescribed in accordance with Method 514, Procedure IX, Part 1, of MIL-STD-810. The equipment shall exhibit no resonance in the frequency range of 10 to 55 Hz that exceed twice the amplitude of the applied vibration. Upon completion of the above vibration tests, the equipment shall meet the requirements of 3.29.4 herein.

4.9.2 Vibration - Part 2 (Vehicular Bounces). - Mast Section AB-1084/TRA-37 and Mast Section AB-1085/TRA-37 shall be subjected separately to the vibration tests prescribed in accordance with Method 514, Procedure IX, Part 2 of MIL-STD-810. Upon completion ofthis test, the equipment shall meet the requirements of 3.29.4 herein.

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4.10 Altitude (Elevation). - The Feedhorn shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810. Prior to, during, and following testing at Altitude, the Feedhorn shall meet full specification performance for the measurements specified in 3.29.5.

4.11 Feedhorn VSWR Test. - The Voltage Standing-Wave Ratio of the Antenna Feedhorn shall be measured by utilizing a standard engineering technique suitable to the contractor after submission to, and approval by the Government. The VSWR shall meet the requirement of 3.28.7.1.

4.12 Feedhorn Coupling Loss (Isolation). - The Feedhorn Isolation (db) shall be measured by utilizing a standard engineering technique suitable to the contractor after submission to, and approval by the Government. The coupling loss shall meet the requirement of 3.28.7.2.

4.13 Feedhorn Watertightness Test. - The feedhorn watertightness test shall be performed as follows: Cap one waveguide port and install a flange pressure adapter on the remaining port. The flange adapter is a UG-406A choke flange with "O"-ring seal modified with a central pipe fitting to allow the entrance of Nitrogen gas. A suitable gas flow meter shall be installed between the gas supply and the pressure adapter. Pressurize the feedhorn to one (1) psig. Test for thirty (30) minutes, recording leakage rate (if any) at ten (10) minute intervals. The feedhorn shall meet the requirements of 3.28.7.3.

4.14 Feedhorn High Power Capability. - The feedhorn high power capability test shall be performed as follows: Set up feedhorn to radiate into a protective absorbing shield. At a selected frequency (preferably 4.7 GHz) within the 4.4 to 5.0 GHz band, increase transmitter power to 2KW $\pm$ 400 watts in 400 watt steps. Monitor the incident and reflected power at each step. A mismatch protection circuit shall be installed and set to trip at a reflected power level of 128 watts (This corresponds to an SWR of 1.65 for a 2KW RF output). Test duration shall be one (1) hour at a level of 2 KW $\pm$ 400 watts of incident power. Incident and reflected power measurements shall be made at ten (10) minute intervals throughout the test cycle. The feedhorn shall meet the requirements of 3.28.7.4 throughout the test.

4.15 Antenna Pattern Measurements. - Antenna patern measurements, including Beamwidth (See 3.28.2) and Minor Lobes (See 3.28.3) shall be conducted at both the Transmit and Receive ports at 4.4, 4.7 and 5.0 GHz. Record both azimuth and elevation patterns at 4.4, 4.7 and 5.0 GHz with the feedhorn oriented, 45 degrees, from a horizontal reference at both Transmit and Receive ports. Record the half-power beamwidth and maximum side lobe levels. Antenna Gain (See 3.28.1) for both Transmit and Receive ports at 4.4, 4.7 and 5.0 GHz shall be recorded. Overall Antenna VSWR

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(See 3.28.4) and Overall Antenna Isolation (See 3.28.5) in the 4.4 to 5.0 GHz range for both the Transmit and Receive ports shall be recorded at this time. A swept frequency measuring and display system or any other standard engineering technique suitable to the contractor after submission to, and approval by the Government may be utilized for these tests.

4.15.1 Transmission Line Attenuation Measurements. - This is normally certified data supplied by the manufacturer of the flexible waveguide sections, however, when data is not supplied, measurement per the requirements of 3.28.6 shall be made at the same time as all other Antenna Pattern measurement testing.

4.15.2 Antenna Test Range. - All tests requiring use of an antenna test range may be performed using any suitable range acceptable to the Government.

4.16 Azimuth Torque. - The antenna azimuth adjustment and cranking torque shall be tested to meet the requirements of 3.28.8. A suitable device shall be employed in order to record the control lever torque.

4.17 Mechanical Interchangeability Inspection. - The equipment shall be tested and inspected to determine conformance to the physical and functional interchangeability requirements of 3.30 as follows:

4.17.1 Procedure:

(a) Select at random one upper mast section, two center mast sections, one lower mast section, one mast base, one erection equipment (less the Baseplate Tie-Back Cable and Truck Assist Erection Cable), two guys lower, two guys upper.

(b) Assemble one lower, center and upper mast section. Make use of the alignment marks and colors to correctly orient the sections.

(c) Remove center mast section and replace with other center mast section.

(d) Secure the mast base to the provided mounting bolts.

(e) Place the assembled mast on the mast base. Note: The mast must lie in line with the arrow on the base plate.

(f) Assemble the upper and lower side guys and remove the slack.

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(g) Assemble the bipod to the mast.

(h) Assemble the gin pole, gin pole stabilizing cable, mast support cable, erection cable and winch.

(i) Take up on winch until the slack is removed from the erection cable.

(j) Select at random two reflector yokes, two space diversity feedhorns, six spars, and two sectionalized reflectors (eight quadrants).

(k) Assemble a reflector yoke to the mast yoke.

(l) Assemble a feedhorn to the reflector yoke.

(m) Remove the feedhorn and the reflector yoke from the mast assembly.

(n) Assemble the other reflector yoke to the mast yoke.

(o) Assemble four reflector quadrants to the reflector yoke.

(p) Remove one quadrant and replace with another quadrant.

(q) Repeat Step P until all quadrants have been mounted.

(r) Assemble a feedhorn to the reflector yoke.

(s) Remove the feedhorn and replace it with the other feedhorn.

(t) Assemble three spars to the feedhorn and reflector. Note: Spars must be afixed to the feedhorn yoke first, then to the reflector.

(u) Remove the spars and replace with other set.

(v) Disassemble and return all parts to their storage area.

4.18 Visual and Mechanical Inspection. - The equipment shall be examined for the acceptable workmanship criteria established by Requirement 9 of MIL-STD-454 and referenced individual parts, materials, and processes specifications and drawings delineated in paragraphs 3.5 through 3.30 of this specification. When parts or materials are not covered by referenced individual specifications, such parts or materials shall be inspected for

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compliance with Specification MIL-P-11268, as follows: They shall be examined for workmanship, mechanical fit, loose nuts, screws, fasteners, rivets, and miscellaneous defects. Controls, fastening devices, and other adjustable or rotatable devices shall be examined for mechanical operation. Soldered connections (on waveguides), grounding connections, brazing, welds, finishes, etc., shall be examined for workmanship. Clearances, dimensions, and mechanical adjustments and fit shall be inspected. Examination also may be made for other visual or mechanical defects, similar to those described above (See 3.31). Visual and Mechanical defects shall be classified by the Government as Major and Minor in accordance with the definitions of MIL-STD-105.

4.19 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5 herein.

4.20 Rough handling test (preparation for delivery). - Rough handling tests per MIL-P-116 shall be conducted as specified in Section 5 when required.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be Level A or C, as specified.

5.1.1 Level A. -

5.1.1.1 Cleaning. - Components of the Antenna Group AN/TRA-37( ), shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Components of the Antenna Group AN/TRA-37( ), shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 and as specified herein.

5.1.1.3.1 Technical literature. - Each Technical manual shall be packaged method IC-1.

5.1.1.3.2 Antenna Group AN/TRA-37.

5.1.1.3.2.1 Components of AN/TRA-37( ). - (When procured less Transit Frames).

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5.1.1.3.2.2 Cables. - All cables shall be wound into a coil of proportionate dimensions and tie in three places with cotton twine conforming to T-T-871.

5.1.1.3.2.3 Stakes. - Stakes shall be bonded together with metal strapping, conforming to QQ-S-781.

5.1.1.3.2.4 Waveguides. - Flexible waveguides shall be wound on th reels provided for the waveguides. Secure the waveguides with the canvas straps on the reel. Place the waveguide covers in poistion and secure the tuumb screws.

5.1.1.3.2.5 Feedhorns. - Each feedhorn shall be individually packaged Method III, as follows; Cushion each feedhorn with blocks or strips of Polystyrene or Polyurethane conforming to PPP-C-850 or MIL-P-26514. Place the cushioned feedhorn within a close-fitting fiberboard box conforming to PPP-B-636, weather-resistant W5C. Box closure shall be as specified in the appendix of the box specification.

5.1.1.3.2.6 Mast Sections. - The upper,center and lower mast sections shall be individually packaged Method III, as specified in 5.1.1.3.2.5.

5.1.1.3.2.7 Transit Assembly. - Each transit shall be individually packaged method III, as follows; Cushion each transit by wrapping in cellulose wadding conforming to PPP-C-843, type 2, Class A. Place the cushioned transit within a close-fitting fiberboard box conforming to PPP-B-636, water-resistant W6c. Box closure shall be in accordance with the appendix of the box specification.

5.1.1.3.2.8 Sledge Hammer, and Spade. - Sledge hammer, and spade, no packaging required.

5.1.1.3.2.9 Reflectors. - No packaging required.

5.1.1.3.2.10 Anchor Assembly. - Anchor assemblies shall be banded together with metal strapping conforming to QQ-S-781. Maximum of four (4) "W" stakes per unit packaging.

5.1.1.3.3 Components of Antenna Group, AN/TRA-37. - (When procured with Transit Frames).

5.1.1.3.3.1 Components. - of the antenna group, shall be placed in position and secured within the Transit Frame. All tie downs or locking devices shall be secured for transit.

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5.1.2 Level C. - Components of the antenna group shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. - Packing shall be Level A, B or C, as specified.

5.2.1 Level A. -

5.2.1.1 Less Transit Frame. - Components of the antenna group packaged as specified in 5.1.1.3.2.1 through 5.1.1.3.2.10, shall be packed within nailed wood boxes conforming to PPP-B-621. Components shall be blocked, braced, and cushioned within the shipping containers in accordance with the applicable requirements of MIL-STD-1186. Shipping containers shall be closed and strapped as specified in the appendix of the box specification.

5.2.1.2 Within Transit Frame. - Components of the antenna group packaged as specified in 5.1.1.3.3.1, requires no further packing.

5.2.2 Level B. - Components of the antenna group packaged as specified in 5.1, shall be packed as specified in 5.2.

5.2.3 Level C. - Components of the antenna group packaged as specified in 5.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of Transportation. (Cushioning shall be provided in accordance with applicable requirements of MIL-STD-1186).

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - Antenna Group AN/TRA-37( ) is intended for use with radio sets employing tropospheric scatter mode of transmission up to 2 kw RF power and space diversity reception in the frequency range of 4.4 to 5.0 GHz.

6.2 Ordering data. - Procurement documents should specify the following:

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(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, Level B, or Level C).

(d) The specific paragraphs of Section 5 which are applicable to the particular procurement.

(e) First Article inspection:

(1) Two (2) sample Antenna Groups AN/TRA-37 (Item Breakout-See Table III) are generally required so that lengthy environmental tests can be completed on one sample unit while complete performance measurements can be made on the second sample unit. (See 3.2)

(2) First Article pack(s) as follows:

Makeup of pack(s).

Number of each kind of pack to be submitted.

Inspection to be performed thereon.

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required.

(i) Quantity of tools and running spare parts required.

(j) Submission of the statement of treatment referenced in 3.16.5, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Color. - The color chip No. X-24087 referenced in 3.11, may be obtained upon request to Commanding General, Aberdeen Proving Grounds, ATTN: Coating and Chemical Laboratory, Aberdeen, Maryland 21005.

6.4 Definitions. -

6.4.1 Equipment divisions. - The terms used herein for equipment divisions conform to Standard MIL-STD-280, entitled "Military Standard Definition of Terms for Equipment Divisions".

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6.4.2 Inspection. - The inspection terms used herein conform to Standard MIL-STD-109, entitled, "Military Standard Quality Assurance Terms and Definitions".

6.5 Group C Inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.6 Nomenclature. - The paratheses ( ) in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/TRA-37-A. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.7 Color of textiles. - Standard samples of Olive-drab No. 7 for fabric and color Shade S-1 for thread referenced in 3.5, may be obtained upon request to Sample Loan Unit, Defense Clothing and Textile Supply Center, Philadelphia, Pennsylvania 19101. When requesting samples, indication should be given as to type of material concerned.

6.8 Verification Inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian

Army - EL

Preparing Activity

Army - EL

Project No. 5820-A458

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-A-55674A(EL), ANTENNA GROUP AN/TRA-37( )

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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